

MAINTENANCE STANDARD OPERATING PROCEDURE
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[FICTIONAL/DEMO DOCUMENT - NOT REAL PROCEDURE]

Document No: SOP-MECH-042 (FICTIONAL)
Title: Centrifugal Pump Maintenance and Inspection Procedure
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Approved By: Chief Engineer (FICTIONAL)

1. PURPOSE

This procedure provides guidelines for inspection, maintenance, and overhaul of centrifugal pumps in process plants. This is a fictional demonstration document for SIH26117.

2. SCOPE

Applicable to all centrifugal pumps in the CDU, VDU, and utility sections.
Equipment classes: P-100 to P-200 series.

3. BEARING INSPECTION AND REPLACEMENT

Section 3.1 - Temperature Limits

- Normal operating temperature: 55-75 degrees Celsius
- Warning threshold: 75-85 degrees Celsius (increase monitoring, schedule inspection)
- Action threshold: Above 85 degrees Celsius (immediate inspection required)
- Critical threshold: Above 95 degrees Celsius (immediate shutdown required)

Section 3.2 - Vibration Limits

- Acceptable: Below 4.5 mm/s RMS
- Warning: 4.5 - 7.0 mm/s RMS (schedule maintenance)
- Action required: Above 7.0 mm/s RMS (maintenance within 48 hours)

Section 3.3 - Bearing Replacement Procedure

- Step 1: Isolate pump from process (close suction/discharge valves, de-energize motor)
- Step 2: Lock-out-tag-out (LOTO) all energy sources
- Step 3: Cool down to ambient temperature (minimum 2 hours after shutdown)
- Step 4: Remove coupling guard and disconnect coupling
- Step 5: Remove bearing housing cover
- Step 6: Extract bearing using appropriate puller (do not hammer)
- Step 7: Inspect shaft for scoring or wear
- Step 8: Install new bearing (clean shaft, correct orientation)
- Step 9: Pack with approved grease (3/4 filled only)
- Step 10: Reassemble and torque all fasteners to specification
- Step 11: Re-align pump-motor coupling (tolerance: +/- 0.05 mm)

Step 12: Run-in period: 2 hours at 50% load, monitor temperature and vibration

4. MECHANICAL SEAL MAINTENANCE

Section 4.1 - Seal Inspection

Inspect for: leakage, heat marks, face damage, spring condition

Acceptable leakage: Less than 1 drop per minute (for standard seals)

Action leakage: 3 or more drops per minute (schedule replacement within 7 days)

Section 4.2 - Seal Replacement Procedure

Step 1: LOTO and isolate pump as per Section 3.3 Steps 1-3

Step 2: Drain pump casing

Step 3: Remove impeller (note: left-hand thread on most pumps in this plant)

Step 4: Remove seal gland plate and spring

Step 5: Slide out rotating seal faces

Step 6: Remove stationary seat from gland

Step 7: Clean all surfaces thoroughly

Step 8: Install new stationary seat (use assembly lubricant)

Step 9: Install new rotating assembly

Step 10: Set correct spring compression per manufacturer spec

Step 11: Reassemble pump

Step 12: Flush seal before startup (flush for 5 minutes minimum)

5. APPROVAL REQUIREMENTS

Section 5.1 - Work Authorization

All maintenance activities on process-critical equipment require:

- Written Work Order signed by Maintenance Supervisor
- Approval from Section Head for planned shutdowns
- Plant Manager approval for unplanned emergency shutdowns
- Safety officer sign-off for hot work

Section 5.2 - Documentation Requirements

- Complete inspection checklist before and after maintenance
- Record all measurements (temperature, vibration, alignment)
- Photograph key components before disassembly
- Update equipment history card

6. SAFETY PRECAUTIONS

- Always use LOTO before working on rotating equipment
- Wear appropriate PPE (safety glasses, gloves, hearing protection)
- Never work on energized equipment
- Ensure area is properly ventilated for hydrocarbon service
- Have spill kit available for oil leaks
- Emergency shower and eyewash within 10 seconds travel

[THIS IS FICTIONAL DEMO DATA]

